

Pediatric

Interpretation of Pediatric Protocols and Basic Principals

The following treatment protocols address the most common illness and injury conditions of childhood. Among ambulance-transported children/minors, the most frequent pre-hospital complaints are trauma, altered mental status (especially seizures), respiratory distress, and toxicological emergencies. Cardiac arrest is the least common pediatric pre-hospital call.

This document defines “children/minor” or “pediatric” patients as less than 15 years for medical conditions and trauma, although physically mature medical patients less than 15 years may be treated in some instances as adults. Other terms include “newborns” (less than 24 hours), “neonates” (less than 28 days) and “infants” (less than 12 months). Pediatric medical patients must be transported to an Emergency Department Approved for Pediatrics (EDAP) facility as stated in Policy 7300.

The protocols are adapted from the Pediatric Prehospital Protocols developed by the Prehospital Treatment Protocol Subcommittee of the Emergency Medical Services for Children (EMSC) Project of the California EMS Authority.

INTERPRETING THE PROTOCOLS

- 1). Basic Life Support (BLS) interventions are outlined in the upper left-hand box of each protocol. These interventions are appropriate for all levels of EMS providers.
- 2). Advanced Life Support (ALS) interventions are outlined in the lower boxes. Within the ALS boxes, the protocols are logically divided internally into one or more treatment pathways, based on key aspects of patient assessment.
- 3). Special considerations for pediatric care are described in the box on the right side of the protocols. Special considerations provide more detailed information about specific elements in the treatment protocols.

- 4). All drug doses are weight-based and given per kilogram. Estimating weight for drug dose calculation is inaccurate and places the patient at risk for under treatment or overtreatment. ***Length-based calculations for drug delivery are more accurate and are preferable in children/minors under age 5 years, when derived from a validated length-based resuscitation tool (e.g., the “First Five Minutes” or “Broselow” tape).*** The Attachment A provides a chart of estimated weights in kilograms for children/minors according to age. The chart also lists age-adjusted vital signs.
- 5). Some critical patients (e.g., multiple system trauma) require rapid hospital transport with minimal on-line communication, and minimal field interventions (e.g., airway and breathing support on scene, with IV attempts enroute). On the other hand, other patients with medical conditions (e.g., status epilepticus, or cardiac arrest) may be appropriately managed in the field prior to ambulance transport.
- 6). Individual presentations may vary because of age-related changes in signs and symptoms, and because of the normal spectrum of acuity of illness and injury. These protocols are not intended to address every permutation or rare situation, and are not intended to replace good clinical judgment.
- 7). ALS interventions in children/minors should ordinarily be minimized, and applied only for appropriate indications. IVs must be used cautiously in children, after carefully weighing the benefits of vascular access against the pain, psychological distress and cost of the procedure.
- 8). Control of pain in children/minors transported by ambulance is an appropriate and important concern. Refer to P-14 Pediatric Pain Management for pain medication doses.

Pediatric Field Primary Survey

PEDIATRIC PRIMARY SURVEY		SPECIAL CONSIDERATIONS			
1.	Establish level of responsiveness.	[1]	Determine scene safety.		
2.	Evaluate airway and protective airway reflexes. [2]	[2]	Identify signs of airway obstruction and respiratory distress, including:		
3.	Basic airway/spinal immobilization PRN. [3] [4]		Cyanosis	Intercostal retractions	Stridor
4.	Oxygen PRN. [5]		Absent Breath Sounds	Drooling	Bradycardia
5.	Assist ventilation PRN. [6][7]		Nasal Flaring	Apnea or Bradypnea	Choking
6.	Stop hemorrhage. Evaluate and support circulation. [8][9]		Tachypnea	Grunting	
7.	Do environmental assessment including consideration of intentional injury. [1]	[3]	Clear airway using suction, jaw thrust and chin lift (and/or head tilt if no suspected spinal trauma). Consider placement of oropharyngeal airway if child unconscious.		
8.	Determine appropriate treatment protocol.	[4]	If cervical spine trauma suspected, immobilize spine with cervical immobilization device. Infants and young children may require under-shoulder support to achieve neutral cervical spine position.		
		[5]	Use nasopharyngeal or oropharyngeal airway, mask, or oxygen by blow-by, as tolerated, with child in position of comfort.		
		[6]	Use chest rise as indicator of adequate ventilation. If chest rise is inadequate, consider: Repositioning the airway Foreign body in airway Inadequate bag volume or activated pop-off valve.		
		[7]	Rescue breathing includes two initial slow breaths (1-1/2 sec) at a rate of 20 breaths per minute for infant or child.		
		[8]	Assess perfusion using: Heart rate Mental Status Skin Signs Capillary refill Blood Pressure Quality of Pulse		
		[9]	Compression rate is at least 100 per minute with 30:2 compression ratio for one person and 15:2 compression ratio for two person. Depths are about 1 ½ inches for infants and about 2 inches for children.		

Pediatric
ATTACHMENT A
WEIGHTS IN KILOGRAMS, AGE-ADJUSTED VITAL SIGNS, FOR CHILDREN

AGE	MEAN WEIGHT in KG	MINIMUM SYSTOLIC BP	NORMAL HEART RATE	NORMAL RESP. RATE	
Premature	<2.5	40	120-170	40-60	
Term	3.5	60	100-170	40-60	
3 months	6.0	60	100-170	30-50	
6 months	8.0	60	100-170	30-50	
1 year	10	72	100-170	30-40	
2 years	13	74	100-160	20-30	
4 years	15	78	80-130	20	
6 years	20	82	70-115	16	
8 years	25	86	70-110	16	
10 years	30	90	60-105	16	
12 years	40	94	60-100	16	

SYSTOLIC BLOOD PRESSURE = 70 + 2 X (age in years)

SOLANO EMERGENCY MEDICAL SERVICES COOPERATIVE

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To: ALL ALS Providers

From: Richard Lotsch, D.O. 
Solano County EMS Medical Director

Date: December 3, 2008

Subject: AICD Magnets

Per Dr Tharratt, the Director of Cal EMS Authority, the use of donut magnets to deactivate an AICD is **NOT** within the Scope of Practice for Paramedics in California.

Effective immediately, donut magnets are to be removed from all ALS units in Solano County, Policy 6606 and Protocol S11 authorizing the use of these magnets for the treatment of a patient with a malfunctioning AICD is hereby **suspended**.

Solano EMS Cooperative

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Glasgow Coma Scale

Eye Opening	Spontaneous			4
	To sound			3
	To pressure			2
	None			1

Verbal Response	Oriented	5	Coos	5
	Confused	4	Cries	4
	Words	3	Screams	3
	Sounds	2	Grunts	2
	None	1	None	1

Motor Response	Obeys Commands / Normal			6
	Localizing			5
	Normal flexion			4
	Abnormal flexion			3
	Extension			2
	None			1

START HERE



Is the patient having a stroke?

✓ Check if yes

Balance

- Perform bilateral index finger-to-nose test and bilateral heel-to-shin test
- Does the patient have sudden loss of balance or coordination, trouble walking or dizziness?

B

Eyes

- Assess 4 quadrants of visual field by having patient locate your index finger.
- Does the patient have trouble seeing out of one or both eyes or sudden double vision?

E

Face

- Ask the patient to smile or show their teeth.
- Does the patient's face look uneven, have sudden drooping or numbness on one side?

F

Arms

- Ask the patient to raise and extend both arms with their palms up.
- Does one arm drift downward?
- Does the patient have sudden numbness or weakness of the arm on one side of the body?

A

Speech

- Ask the patient to say, "You can't teach an old dog new tricks".
- Does the patient have slurred speech, trouble speaking, understanding or seem confused?

S

Time

- What time did the symptoms start? _____:
- What time was the patient last known well (last appear normal)? _____:

T

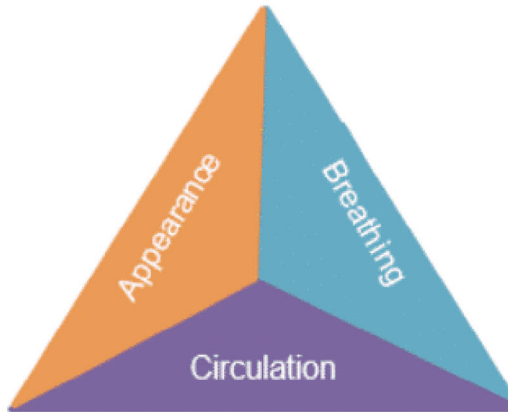
Glucose level = _____

APGAR Score

Appearance	2 Completely pink	1 Extremities blue	0 Completely blue/pale
Pulse	2 Over 100/min	1 Under 100/min	0 No pulse
Grimace	2 Cries or coughs	1 Grimaces or weak cry	0 No response
Activity	2 Active movement	1 Arms/Legs flexed	0 No movement
Respiration	2 Strong cry	1 Slow or irregular	0 No breathing

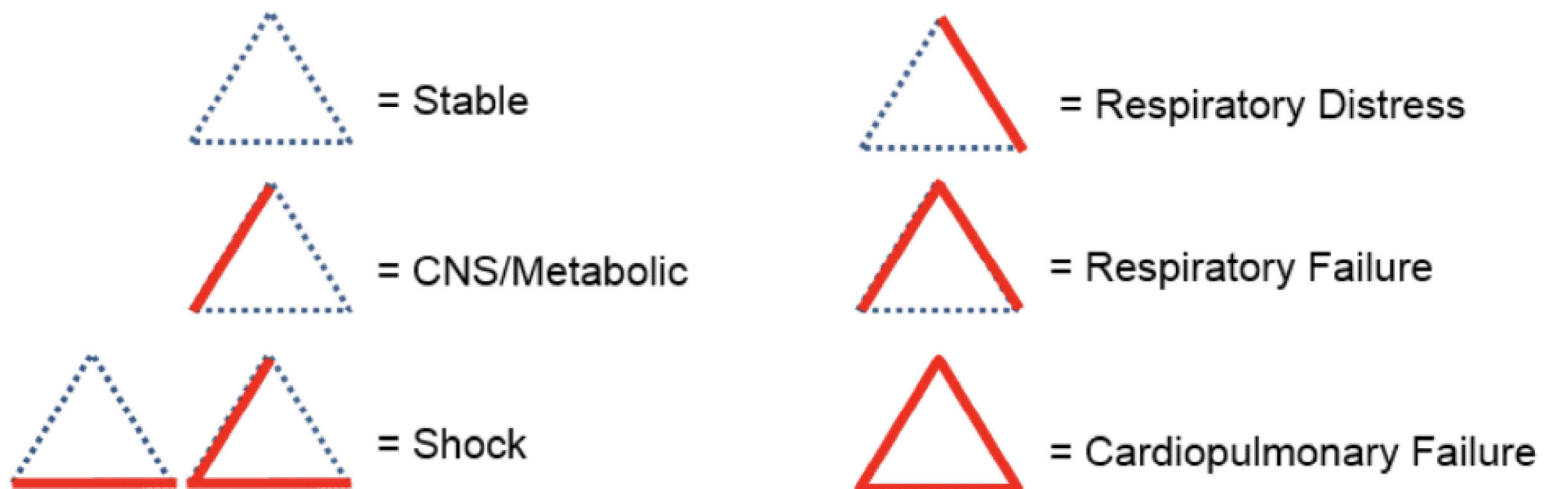
First Impression

Appearance Triage
Unresponsive, Limp/Apathetic, Inconsolable, Weak or hoarse cry
Mild distress, Intermittently consolable
Alert, Tracking with eyes, Strong muscles, Appropriate interactions (may include crying)



Breathing Triage
Central cyanosis, Absent or labored breathing, Obstructed (stridor, blood/vomit/foreign body)
Excess secretions, Wheezing, Accessory muscle use
Normal respiratory effort and rate

Circulation Triage
Pallid or mottled, Cyanotic
Mild delay in capillary refill or cool digits
Pink, Normal



Pediatric Average Normal Weights and Vital Signs

Age Group	Weight (lbs)	(kg)	Pulse	Systolic BP	Respirations
Premature	3	1.5	100-180	50-60	40-60
Newborn	7.5	3.5	100-160	60-90	40-60
1 year	22	10	110-170	70-110	20-30
3 years	33	15	80-160	80-110	20-30
6 years	44	20	60-130	90-115	20-30
8 years	55	25	60-120	90-115	12-25
12 years	88	40	60-120	95-120	12-25
15 years	125	55	60-120	100-130	12-20